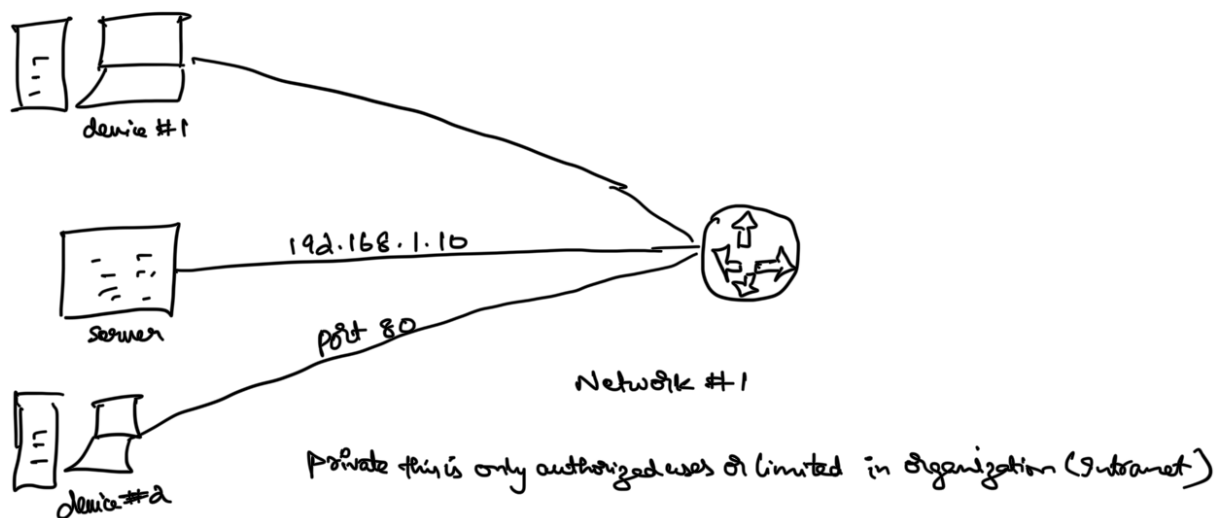


Extending your Network : —

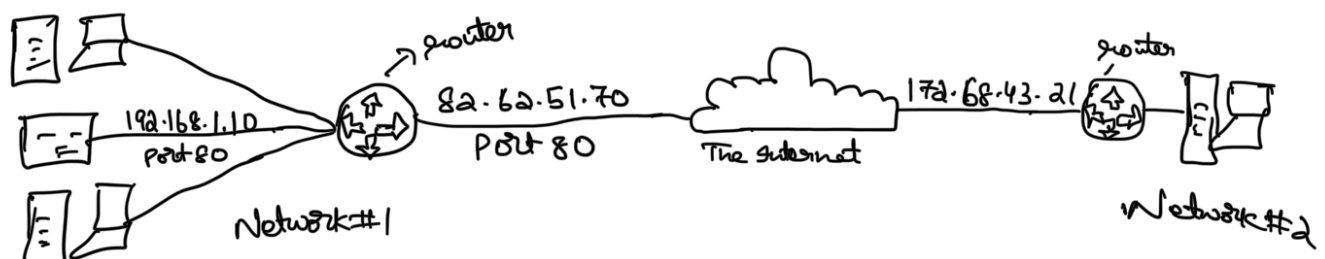
Task 1:- Introduction to port forwarding

port forwarding is essential component in connecting applications and services to the Internet. without port forwarding, applications and services such as web servers are only available to devices within same network.

Ex:- The network, the server with an IP address "192.168.1.10" runs a webserver on port 80. only two other computers on this network will be able to access it (this is known as Intranet).



If the administrator wanted website to be accessible to the public (using the Internet), they would have to implement port forward, like this diagram below :



with this design, Network #2 will now be able to access the webserver running on

Network #1 using public IP address of Network #1 (82.69.51.70)

* It is easy to confuse port forwarding with behaviour of firewall. However we just understand port forwarding opens specific ports (recall how packet works). In comparison, firewalls determine if traffic can travel across these ports (even if these ports are open by port forwarding).

~~***~~ * port forwarding is configured at the router of a network.

Quiz:-

1) what is name of device that is used to configure port forwarding?

A) Router.

Task 2:- Firewalls 101:-

Note:- A Firewall is a security tool, hardware or software that is used to filter network traffic by stopping unauthorized incoming and outgoing traffic.

* A firewall is a device within network responsible for determining what traffic is allowed to enter or exist. Think of firewall as border security for a network.

An administrator can configure firewall to permit or deny traffic from entering or existing network based on numerous factors such as:

* where the traffic is coming from? (has the firewall been told to accept/deny traffic from specific network?)

* where the traffic going to? (has the firewall been told to accept/deny traffic destined for specific network?)

* what port is the traffic for? (has firewall been told to accept/deny traffic destined for port 80 only?)

* what protocol is the traffic using? (has firewall been told to accept/deny traffic that is UDP, TCP or both?)

Firewalls perform packet inspection to determine the answer of these questions.

Firewalls comes in all shapes and sizes. From dedicated pieces of hardware that can handle magnitude of data to residential routers. Firewalls can be categorize into 2 to 5 categories.

Firewall |

Category	Description
Stateful	* This type of firewall uses entire information from a connection, rather than inspecting an individual packet, this firewall determine behaviour of a device based upon entire connection. * This firewall type consumes many resources in comparison to stateless firewalls as decision making is dynamic. * If a connection from host is bad, it will block the entire device.
Stateless	* This type of firewall uses static set of rules to determine whether or not individual packets are acceptable or not. - Ex:- If device sending bad packet will not necessarily mean the entire device is then blocked. * However this firewalls are great when receiving large amount of traffic from set of hosts

Quiz:-

Q) What layers of OSI model do firewalls operate at?

A) 3 & 4 → (Networking & transport)

Q) What category of firewall inspects the entire connection?

A) Stateful

Q) What category of firewall inspects individual packet?

A) Stateless